A Privacy Preserving Warehouse Coordination System for National Paddy Distribution Using Zero Knowledge Proofs, Graph Attention Networks and Permissioned Blockchain

Buwaneka Vishwajith
Department of Software Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it22372444@my.sliit.lk

Sasin Ransara
Department of Software Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it22102096@my.sliit.lk

Ishara Chamudi
Department of Software Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it22147950@my.sliit.lk

Neranda Dilhara
Department of Software Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it22294784@my.sliit.lk

Mahima Weerasinghe
Department of Computer Science
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
mahima.w@sliit.lk

Eishan Weerasinghe
Department of Software Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
eishan.w@sliit.lk

***Abstract*- Sri Lanka's Paddy Marketing Board (PMB) coordinates paddy warehouse operations through informal communication channels lacking verifiable data exchange and accountability. This paper presents a privacy-preserving coordination system addressing four critical gaps: the absence of tamper-evident stock reporting, unverifiable capacity claims during disaster response, undetected anomalous warehouse behaviour, and non-auditable redistribution decision trails. First, a Zero-Knowledge Proof circuit implemented using the Groth16 proving scheme enables supervisors to prove that a reported storage capacity meets a required threshold without that value being written to the shared ledger or disclosed to peer organisations. Second, a Graph Attention Network trained on behavioural features including damage ratios, inflow patterns, utilisation rates and redistribution frequencies computes per-warehouse reliability scores by exploiting relational signals across the warehouse network graph. Third, a permissioned Hyperledger Fabric blockchain on a two-organisation channel anchors stock events, disaster declarations, redistribution orders and proof verification results as immutable records, providing a timestamped audit trail persisting independently of the application database. Prototype evaluation indicates the circuit enforces capacity constraints with zero false acceptances; the Graph Attention Network correctly classified 26 of 30 nodes under in-sample evaluation on a synthetic dataset with a 24-node majority class; and blockchain anchoring provides tamper detection confirmed through SHA-256 hash verification. The system is presented as a proof-of-concept prototype demonstrating the architectural viability of combining these three mechanisms in a government agricultural logistics context.**

***Keywords*- zero knowledge proofs, graph attention networks, Hyperledger Fabric, blockchain, paddy warehouse, anomaly detection, smart contracts, SHA-256**

I. INTRODUCTION

The Paddy Marketing Board (PMB) is the central government body responsible for purchasing, storing and distributing rice through a dispersed network of regional warehouses. Rice is Sri Lanka's main staple food, and national food security depends on this network's operational integrity, particularly during seasonal surplus, natural disasters and regional supply imbalance. Despite this significance, PMB warehouse coordination still relies on unofficial channels such as phone calls and mobile messaging for reporting stock movements, announcing disasters and organising inter-warehouse redistribution, as reported by a serving PMB Regional Manager consulted during requirements gathering for this study. The PMB's mandate and network structure are documented in [1]; current coordination practice is not.

This dependency introduces four compounding vulnerabilities. Supervisor stock event reports lack cryptographic verification, making fraudulent quantity reporting undetectable by later audit. A warehouse cannot demonstrate that its available capacity meets a threshold without revealing its precise stock level, so Regional Managers must select redistribution sources from self-reported figures alone. The network lacks any systematic method for identifying unusual behavioural trends such as persistently high damage claim rates or implausible inflow-to-utilisation disparities. And redistribution decisions are not recorded in any immutable, independently verifiable form, leaving post-hoc auditing dependent on informal records susceptible to alteration or loss.

Previous research has addressed individual facets of this problem blockchain-based agricultural traceability [2], [3], zero-knowledge proofs for ledger privacy [4], and graph neural network anomaly detection in transaction networks [5] but no system combines them within a single warehouse coordination architecture. This paper addresses all four gaps

through three contributions: a Circom zero-knowledge circuit using Groth16, letting a supervisor prove that a reported capacity meets a minimum threshold without revealing the value; a two-layer Graph Attention Network with four attention heads, computing per-warehouse reliability ratings from behavioural signals propagated across the redistribution graph; and a permissioned Hyperledger Fabric blockchain on a two-organisation channel, anchoring operational events as immutable records queryable independently of the application database. A composite rating formula integrates the three:

$$\mathbf{S = 0.40 \cdot d + 0.35 \cdot c + 0.25 \cdot r} \quad (1)$$

where d is the normalised inverse distance score, c is the normalised available capacity score, r is the GNN-derived reliability score for each candidate warehouse, and S is the composite selection score.

The weights were assigned by the authors to reflect operational priority in disaster logistics: distance highest, because delivery time is the binding constraint during emergency response; capacity second, because a proximate warehouse without room cannot fulfil an order; and reliability third, as a modifier discriminating between logistically comparable candidates rather than overriding physical constraints. These values are not empirically optimised and no sensitivity analysis was conducted.

II. Related Studies

A. Blockchain-Based Agricultural Traceability and Coordination

Blockchain has been widely applied to agricultural supply chain transparency for tamper-evident record keeping and provenance verification. Tian [2] combined RFID with a permissioned blockchain for food safety traceability, showing how immutable records remove the single points of failure of centralised agricultural databases. Mondal et al. [3] extended this by binding physical items to ledger records through tag readings a pattern of coupling a custody event to an immutable digital record that the present work applies to warehouse stock events.

Warehouse management in agricultural settings has received comparatively little attention. Kamble et al. [6] observe that research concentrates on traceability, provenance and production analytics while downstream logistics coordination receives far less coverage, a gap confirmed by Yogarajan et al. [14]; blockchain work in humanitarian logistics has likewise concentrated on donation and fund traceability rather than inter-facility stock redistribution [19]. Cryptographic audit trail mechanisms for redistribution decisions are, to the authors' knowledge, absent from this literature, a gap the present work closes by anchoring disaster declarations, redistribution orders and proof verification results as chaincode-managed assets, permitting audit queries independent of the application database.

B. Zero-Knowledge Proof Applications in Supply Chain Privacy

Ben-Sasson et al. [7] established the practicality of succinct non-interactive arguments of knowledge, laying the groundwork for Groth16 [8], used here, which delivers constant-size proofs of three group elements verifiable through a single pairing product equation. Narula et al. [4] applied zero-knowledge balance proofs to distributed-ledger auditing, confirming aggregate properties of private holdings without revealing them. Choudhuri et al. [9] establish that a blockchain can serve as a public setup for secure computation. The present work applies these principles through a GreaterEqThan(32) circuit proving $x \geq$ threshold without revealing x, producing a three-signal proof verifiable on-chain through Fabric chaincode.

C. Graph Neural Network Anomaly Detection

Graph Neural Networks perform well in anomaly detection where relational structure supplies discriminative information unavailable from node-level characteristics alone. Kipf and Welling [10] presented the Graph Convolutional Network formulation, extended by Veličković et al. [11] with attention mechanisms that learn adaptive edge weights during message passing relevant here, since the significance of a redistribution link is not uniform across the network.

Dou et al. [5] showed that GNN-based fraud detection in financial transaction graphs significantly outperforms feature-based classifiers by capturing collusive patterns across connected accounts, directly analogous to the collusive redistribution pattern addressed here. The attention-based design is further motivated by Liu et al. [12], who show that fraud detection GNNs degrade when neighbour relevance is inconsistent across a graph, supporting adaptive rather than uniform neighbour weighting.

D. Integrated Smart Contract and AI Coordination Frameworks

Salah et al. [13] identify separation between off-chain inference and on-chain result anchoring as the appropriate pattern for production deployment. The same separation is followed here: inference runs in an isolated Python service, while reliability scores are persisted relationally and referenced during ranking rather than anchored individually.

A comparison of the reviewed approaches against the capabilities required for the present problem is given in Table I. To the best of the authors' knowledge, no previous system combines permissioned blockchain audit trails, graph-structured anomaly detection and zero-knowledge capacity verification within a single agricultural warehouse coordination architecture; the present study fills this gap for Sri Lanka's national paddy distribution network.

TABLE I. Capability Comparison of Related Approaches

Work	Immutable audit trail	ZK capacity proof	Graph anomaly detection	Warehouse coordination	Disaster response
Tian [2]	Yes	No	No	No	No
Narula [4]	Yes	Yes	No	No	No
Dou [5]	No	No	Yes	No	No
Salah [13]	Yes	No	No	No	No
Yogarajan [14]	Yes	No	No	Partial	No
Negi [19]	Yes	No	No	No	Yes
This work	Yes	Yes	Yes	Yes	Yes

a. Capabilities are assessed against the requirements of the present problem, not against the objectives of the cited works.

III. Methodology

A. System Architecture Overview

The system follows a microservice-aligned layered architecture in which each contribution runs as an independent service orchestrated by a central Node.js backend, implemented in TypeScript with Express and connected to a MySQL database through an object-relational mapping layer. All client-facing API endpoints are

authenticated using JSON Web Token pairs, comprising a short-lived access token and a long-lived refresh token with rotation and replay detection, and authorised through role-based middleware enforcing four roles: Administrator, Regional Manager, Warehouse Supervisor and Auditor.

Hyperledger Fabric was chosen over public alternatives for deterministic transaction finality, a permissioned identity model aligning with the government organisational hierarchy, and endorsement policies enabling multi-organisation consensus without proof-of-work overhead. The network is deployed as a two-organisation channel using the test-network configuration with CouchDB as world state database, required for the rich JSON queries used in warehouse history and audit trail retrieval.

B. Blockchain Chaincode Design

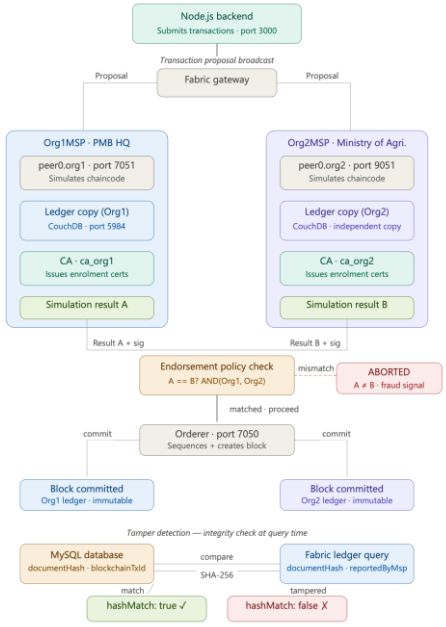


Fig. 1. Hyperledger Fabric transaction endorsement and commit flow across the two-organisation channel, with the SHA-256 hash comparison used for document tamper detection at query time.

The chaincode defines asset types for stock events, disaster events, redistribution orders and proof verification results, each persisted to the CouchDB world state under a type-prefixed key enabling selector queries without full ledger scans, and exposes functions for recording each type and querying warehouse history and disaster audit trails. The redistribution order function additionally computes an integrity digest as the SHA-256 hash of the concatenated order fields, letting auditors verify that parameters were not altered off-chain after anchoring. It is not a digital signature and provides no non-repudiation: any party reconstructing the same field values produces the same digest. Attribution is therefore layered the submitting organisation is identified on-chain through the membership service provider identity from the transaction context, the individual user only relationally. Per-user non-repudiation would require enrolment certificates issued by the organisation's certificate authority, identified as future work.

The backend connects to the Fabric peer through the gateway SDK using a singleton gateway pattern, authenticating with Admin identity certificates from the Org1 membership service provider and TLS verification against the peer's certificate authority. Submit transactions invoke chaincode write functions and await orderer commitment; evaluate transactions invoke read functions without creating on-chain records. The endorsement and commit flow across the two-organisation channel is shown in Fig. 1.

C. Zero-Knowledge Proof Pipeline

The second gap is filled by the zero-knowledge proof component, letting a warehouse demonstrate that available capacity meets a disaster response threshold without revealing precise stock levels. The circuit is written in Circom 2.2.3 using the GreaterEqThan(32) comparator from circomlib, which constrains the comparison to values representable in 32 bits and therefore supports values far exceeding any realistic warehouse capacity.

The circuit defines one private input, availableCapacity, and two public inputs, threshold and warehouseCommitment. The output signal valid is constrained to equal 1, so proof generation fails at the circuit level if availableCapacity is less than threshold. This is a critical security property: a dishonest warehouse cannot generate a valid proof for capacity it does not possess, since the constraint is enforced by the Rank-1 Constraint System before the proving key is applied. The three public signals are valid, threshold and warehouseCommitment; availableCapacity appears in no public output.

The warehouseCommitment signal is declared as a public input but is not constrained within the circuit in the current implementation. Warehouse identity is instead bound at the application layer, the submission endpoint associating each verified proof with the warehouseId and disasterEventId of the authenticated request and rejecting duplicates. This is a limitation: the proof does not attest to which warehouse produced it, leaving that binding dependent on the application server rather than the proof system. A circuit-level constraint of the form Poseidon(warehouseId, availableCapacity, salt) === warehouseCommitment, with warehouseId and salt as additional private inputs, would relocate the guarantee into the proof, Poseidon being preferred over an in-circuit SHA-256 costing tens of thousands of constraints against a few hundred. This is identified as immediate future work.

The trusted setup follows the Groth16 two-phase ceremony: Powers of Tau at depth 14, followed by a circuit-specific phase 2 contribution producing the proving key, with the verification key loaded once at startup. Proof generation is currently server-side; the production architecture intends browser-side generation using the compiled WASM artifact. On successful verification the result is persisted relationally and anchored on the ledger, which stores SHA-256 hashes of the proof and its public signals rather than the full proof data.

1) Threat Model and Privacy Scope: The guarantee does not hold uniformly against all parties. The application server learns availableCapacity, since proof generation executes server-side, and the Regional Manager observes capacity through the term c in (1); neither is treated as an adversary here. The value is never written to the ledger and appears in no public signal, so it is not disclosed to the second endorsing organisation, to peer warehouses, or to any auditor querying the ledger. The property claimed is therefore confidentiality with respect to the shared ledger and to other organisations, not with respect to the application server.

The circuit attests that the supplied value satisfies the threshold, not that it corresponds to physical inventory; establishing that would require the private input to derive

from an authenticated record, for example a Merkle inclusion proof against a periodically anchored stock-ledger root verified in-circuit. The prototype therefore prevents disclosure of the reported capacity but does not prevent a supervisor from reporting an inflated figure. Detection of that class of misreporting is delegated to the anomaly component of Section III-D, which operates over reporting patterns rather than individual claims. The two mechanisms are consequently complementary: the proof addresses disclosure, the anomaly detector truthfulness.

2) Mitigation of Threshold-Probing Leakage: Because threshold is public, an adversary able to request proofs at chosen thresholds could recover the capacity by binary search in O(log n) queries. Two mitigations apply: one accepted proof per warehouse and disaster pair, with duplicates rejected; and rate limiting on all authenticated endpoints. Restricting thresholds to fixed capacity bands, not implemented here, would bound disclosure to the band boundaries regardless of query volume.

D. Graph Attention Network for Reliability Scoring

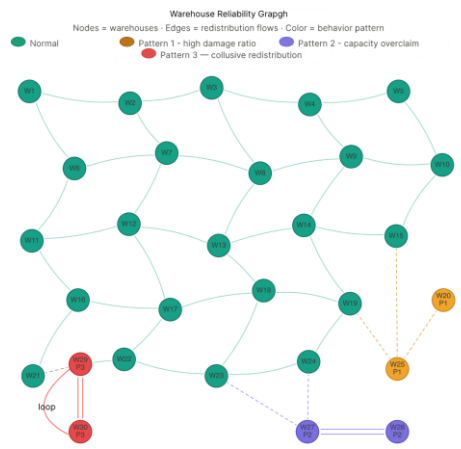


Fig. 2. Synthetic warehouse graph used for GAT training and evaluation. Nodes are warehouses, edges are historical redistribution flows, and colour indicates the injected behavioural pattern.

The GNN component addresses the third gap: detection of anomalous behavioural patterns that individual metric thresholds cannot identify in isolation. The system constructs a homogeneous graph in which nodes represent active warehouses and directed edges represent historical redistribution orders, with node features computed from live warehouse, stock event and redistribution records into the eight-dimensional vector defined in Table II.

TABLE II. WAREHOUSE NODE FEATURE VECTOR DEFINITION

Index	Feature Name	Formula
f0	total_events	min(count / 100, 1.0)
f1	inflow_ratio	inflow_events / total_events
f2	damage_ratio	damage_events / total_events
f3	utilization_pct	current_stock / capacity
f4	avg_quantity	min(avg_tons_per_txn / 500, 1.0)
f5	event_frequency	min((total_events / days_active) / 10, 1.0)
f6	has_disaster	1.0 if active disaster else 0.0
f7	redistribution_count	min(outgoing_orders / 10, 1.0)

a. current_stock = Σinflow_tons − Σoutflow_tons, using the same formula as the backend stock service. days_active = max(1, today − warehouse creation date), preventing division by zero.

b. Normalisation denominators are order-of-magnitude scaling constants mapping each feature into approximately [0,1]. They were not tuned against operational data and would require recalibration on real data.

Three anomaly patterns are detectable from these features in combination, corresponding to the classes injected into the synthetic dataset. Excessive damage claiming appears as f2 in the range 0.3 to 0.7; capacity overclaiming as high f1 with near-zero f3, indicating stock reported as received and simultaneously as lost without ever being held; and collusive redistribution as f7 between 0.7 and 1.0 with elevated f5, suggesting coordinated transfers inconsistent with legitimate emergency response. The remaining features are contextual: f0 identifies dormant warehouses, f4 flags implausible volumes, and f6 prevents false penalisation of warehouses legitimately in disaster response. These ranges are configuration parameters of the data generator chosen to produce a separable synthetic signal, reported for reproducibility rather than as empirically grounded thresholds; empirical loss distributions were unavailable for the PMB network.

The model comprises two graph attention convolution layers. The first maps the eight-dimensional input to sixteen dimensions across four attention heads, giving a sixty-four-dimensional concatenated output; the second reduces this to an eight-dimensional embedding with a single head. A two-layer perceptron with ReLU and dropout (p = 0.3) maps each embedding to a reliability score in [0, 1] through a sigmoid unit. The attention coefficient for edge (i, j) is computed as:

$$\alpha_{ij} = \mathrm{softmax}(\mathrm{LeakyReLU}(a^{T} [Wh_i \,\|\, Wh_j])) \qquad (2)$$

where **W** is a shared linear transformation matrix, **a** is a learnable attention vector, h_i and h_j are the feature vectors of nodes *i* and *j* respectively, and || denotes vector concatenation [11]. Each node's updated representation is then computed as the weighted sum of its transformed neighbours:

$$h'_i = \mathrm{ELU}(\Sigma_{j \in N(i)} \alpha_{ij} \cdot Wh_j) \qquad (3)$$

where N(i) includes node *i* itself through self-loops added at each attention layer.

The justification for graph-structured modelling over per-warehouse classification is that a node's representation is influenced by its neighbours' behavioural profiles. Consider an established Warehouse X with a redistribution edge to a newly opened Warehouse Y exhibiting the capacity-overclaiming pattern. After applying the attention mechanism X attends to Y with weight α = 0.431, blending 43.1 percent of Y's transformed representation into X's embedding; X's reliability score falls from 0.625 in isolation to 0.531, while Y scores 0.415, below the 0.5 threshold. A warehouse's redistribution connections therefore influence its score in addition to its own reported figures. This is an illustrative trace of model behaviour on two nodes, not an experimental comparison against non-relational classifiers; no such comparison was conducted, as noted in Section V. The model is trained on a synthetic dataset of 30 warehouse nodes, 24 labelled reliable and 6 anomalous, generated with fixed random seeds (NumPy 42, PyTorch 42) for reproducibility. The resulting graph structure and injected behavioural patterns are shown in Fig. 2. Training runs for 200 epochs using the Adam optimiser (learning rate 0.01, weight decay 5×10^{-4}) minimising binary cross-entropy loss over all nodes simultaneously. At inference the system reads live warehouse and redistribution records, constructs the current graph and passes it to the trained model; a dedicated model service exposes an endpoint returning per-warehouse reliability scores, which feed into (1).

E. Disaster Response Coordination and Document Integrity

When a Regional Manager reports a disaster, the system uses (1) to score each candidate warehouse, combining the most recent reliability score, the normalised available capacity and the geographic distance computed with the Haversine formula. The proof verification flag is also exposed in the ranked list, letting the Regional Manager distinguish warehouses that have merely self-reported capacity from those that have cryptographically demonstrated it.

The full disaster lifecycle, comprising declaration, proof verification, redistribution ordering and resolution, is recorded in a chronological audit timeline exposed through a dedicated endpoint, assembled from relational records combined with on-chain assets retrieved via chaincode evaluate transactions, giving auditors a single view cross-referencing off-chain data against independently verifiable blockchain records.

A React Native mobile client presents a distinct interface per role, supporting stock event recording and document upload for Supervisors, warehouse and user administration for Administrators, disaster management and redistribution for Regional Managers, and read-only audit and tamper verification for Auditors.

IV. Results and Discussion

A. Zero-Knowledge Proof Circuit Evaluation

The circuit was evaluated for constraint enforcement and verification accuracy. Proof generation fails at the circuit level whenever availableCapacity is less than threshold, the constraint on the valid signal producing an unsatisfiable witness assignment before the prover is invoked; zero false acceptances were recorded across all invalid inputs. At equality, generation succeeded and verification returned a positive result. Duplicate submissions for the same warehouse and disaster pair were rejected with HTTP 409, preventing replay. Generation took approximately two to five seconds server-side.

The zero-knowledge property is not established empirically; it follows from the security of Groth16, proven zero-knowledge under pairing-based assumptions in the generic group model [8], conditional on honest trusted setup. Each proof intentionally reveals the threshold, the commitment signal, and that the constraint was satisfied. Inspection of the public signals confirmed no additional value was exposed through the circuit's output interface, which verifies correct circuit construction rather than the cryptographic property itself.

B. Graph Attention Network Evaluation

The GAT model was evaluated on the synthetic dataset of 30 warehouse nodes with eight behavioural features per node, trained for 200 epochs. Binary cross-entropy loss decreased from 0.4930 at epoch 50 to 0.3767 at epoch 100 and 0.3745 at epoch 150, reaching 0.2989 at epoch 200, with 26 of the 30 nodes correctly classified.

This figure requires careful interpretation. The evaluation is in-sample, measuring fit rather than generalisation, and the class distribution is imbalanced at 24 reliable to 6 anomalous, so a majority-class classifier would attain 24 of 30; the result is an improvement of two nodes over that baseline. Per-class precision and recall were not recorded, so performance on the anomalous class the class of operational interest is unknown, and overall accuracy on an imbalanced set does not constrain it. The four errors are consistent with feature-distribution overlap at the boundaries of the injected ranges, though unverified. The result should be read as evidence that training converges and the injected patterns are learnable, not as evidence of detection performance; the evaluation design required for a performance claim is set out in Section V.

C. Blockchain Audit Trail Evaluation

On-chain record integrity was verified by querying the chaincode directly through the peer command-line interface, bypassing the application backend; all anchored assets were retrieved correctly, confirming that records cannot be altered through the relational database alone. Document tamper detection was confirmed by re-uploading an original stock report to the verification endpoint, which returned a positive result, while the same document modified by a single character returned a negative one. The document hash in the corresponding on-chain asset remained unchanged, providing tamper evidence independent of the application layer.

An endorsement consistency issue caused by non-deterministic timestamp generation was resolved by deriving all timestamps from the ledger-supplied transaction context, identical across endorsing peers. The full disaster audit trail was retrieved in correct chronological order, all entries carrying verified transaction identifier references.

These results demonstrate tamper evidence under the trust assumptions of the deployed configuration rather than absolute immutability. The endorsement policy prevents unilateral alteration by either organisation, but collusion between both, compromise of a certificate authority signing key, or administrative filesystem access to both peers would each defeat the guarantee, and the single-node ordering service is not representative of a production Raft deployment. Governance arrangements determining which institutions operate each organisation lie outside the technical scope of this study.

The evaluation is qualitative; the performance and fault behaviour measurements not conducted are listed in Section V, for which established Fabric benchmarking methodologies [17], [18] provide the basis.

V. Limitations and Threats to Validity

Construct validity. Anomaly ranges, normalisation constants and graph topology were chosen by the authors to produce a separable signal, encoding hypotheses about the form fraudulent behaviour would take; whether real PMB misreporting exhibits these signatures is unverified. Graph edges are randomly generated and do not reproduce realistic redistribution topology, particularly limiting confidence in the collusive-redistribution pattern, of which the dataset contains few instances.

Internal validity. The classification result is in-sample and must be read against a majority-class baseline of 80 percent. Per-class precision and recall were not recorded, so performance on the anomalous class is unknown. A stratified partition repeated across multiple seeds, reporting per-class precision, recall, F1 and precision–recall AUC with a confusion matrix, is required before any performance claim, as is comparison against non-relational baselines and an edge-ablated variant. Neither was conducted, so the claim that

graph structure confers an advantage rests on results for analogous tasks in the literature [5], [12], [15] rather than on experiment here.

External validity. Evaluation used five operational warehouses and a 30-node synthetic graph. Behaviour at full PMB network scale is unrepresented, and no stakeholder evaluation was conducted, so the operational usefulness of the reliability score is untested.

Cryptographic scope. Proof generation executes server-side, so the server observes the private value; confidentiality holds with respect to the shared ledger and other organisations, not the server. The circuit binds the proof neither to a warehouse identity nor to physical inventory. Groth16 requires a circuit-specific trusted setup, performed here by a single party; retention of the setup randomness would permit forgery, and deployment requires a multi-party ceremony. The construction is not post-quantum secure. Timing was recorded only as an aggregate range on one device; systematic benchmarking of the kind reported in [16] was not performed.

Blockchain and end-to-end scope. The deployment uses a two-organisation test network with a single-node ordering service on one host; latency, throughput, storage growth, failure behaviour, and conflicting or unauthorised transaction handling were not measured. No instrumented measurement of a complete disaster-response workflow was conducted, so completion time, failure rate and required human actions are unknown.

Reproducibility. Seeds are fixed (NumPy 42, PyTorch 42), so training behaviour is reproducible. Circuit source, data generation scripts, chaincode and configuration are available from the authors on request.

VI. Conclusion

Evaluation confirmed that the proof circuit enforces capacity thresholds with zero false acceptances, rejecting invalid claims before proof generation, and that audit trail integrity holds under direct chaincode queries bypassing the application layer. The anomaly detector's in-sample result establishes that training converges and the injected patterns are learnable; whether the graph-structured formulation outperforms feature-only classifiers remains open.

Three evaluation gaps identified in Section V require closure first: a stratified partitioned evaluation with per-class metrics and baseline comparison, systematic performance characterisation of the proof and blockchain components, and an instrumented end-to-end measurement. Four directions follow: retraining the GAT on real PMB transaction data with a formal train, validation and test split, the critical step for establishing generalisation; migrating proof generation to browser-side WASM to preserve the zero-knowledge property with respect to the server; scaling to a multi-orderer Raft configuration with organisations representing PMB regional offices; and constraining the warehouse commitment within the circuit while deriving the private input from an authenticated stock record, extending the proof's guarantee from the reported value to the recorded one.

The system demonstrates that simultaneous integration of privacy-preserving cryptographic proofs, graph-structured machine learning and permissioned blockchain audit trails is architecturally viable in a government agricultural logistics context. It is presented as a proof-of-concept prototype; establishing deployment suitability would require real PMB transaction data, a production-scale Fabric configuration and structured stakeholder assessment, none of which was within scope.

References

[1] Paddy Marketing Board, "Annual report and accounts," Ministry of Agriculture, Sri Lanka, 2023. [Online]. Available: https://www.pmb.gov.lk. [Accessed: Aug. 22, 2026].

[2] F. Tian, "A supply chain traceability system for food safety based on HACCP, blockchain and Internet of Things," in Proc. 2017 Int. Conf. Service Systems and Service Management (ICSSSM), Dalian, China, Jun. 2017, pp. 1–6, doi: 10.1109/ICSSSM.2017.7996119.

[3] S. Mondal, K. P. Wijewardena, S. Karuppuswami, N. Kriti, D. Kumar, and P. Chahal, "Blockchain inspired RFID-based information architecture for food supply chain," IEEE Internet Things J., vol. 6, no. 3, pp. 5803–5813, Jun. 2019, doi: 10.1109/JIOT.2019.2897527.

[4] N. Narula, W. Vasquez, and M. Virza, "zkLedger: Privacy-preserving auditing for distributed ledgers," in Proc. 15th USENIX Symp. Networked Systems Design and Implementation (NSDI), Renton, WA, USA, Apr. 2018, pp. 65–80.

[5] Y. Dou, Z. Liu, L. Sun, Y. Deng, H. Peng, and P. S. Yu, "Enhancing graph neural network-based fraud detectors against camouflaged fraudsters," in Proc. 29th ACM Int. Conf. Information and Knowledge Management (CIKM), Virtual Event, Ireland, Oct. 2020, pp. 315–324, doi: 10.1145/3340531.3411903.

[6] S. S. Kamble, A. Gunasekaran, and S. A. Gawankar, "Achieving sustainable performance in a data-driven agriculture supply chain: A review for research and applications," Int. J. Production Economics, vol. 219, pp. 179–194, Jan. 2020, doi: 10.1016/j.ijpe.2019.05.022.

[7] E. Ben-Sasson, A. Chiesa, E. Tromer, and M. Virza, "Succinct non-interactive zero knowledge for a von Neumann architecture," in Proc. 23rd USENIX Security Symp., San Diego, CA, USA, Aug. 2014, pp. 781–796.

[8] J. Groth, "On the size of pairing-based non-interactive arguments," in Advances in Cryptology—EUROCRYPT 2016, Lecture Notes in Computer Science, vol. 9666, M. Fischlin and J.-S. Coron, Eds. Berlin, Germany: Springer, 2016, pp. 305–326, doi: 10.1007/978-3-662-49896-5_11.

[9] A. R. Choudhuri, M. Green, A. Jain, G. Kaptchuk, and I. Miers, "Founding secure computation on blockchains," in Advances in Cryptology—EUROCRYPT 2019, Lecture Notes in Computer Science, vol. 11478, Y. Ishai and V. Rijmen, Eds. Cham, Switzerland: Springer, 2019, pp. 351–380, doi: 10.1007/978-3-030-17659-4_13.

[10] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in Proc. 5th Int. Conf. Learning Representations (ICLR), Toulon, France, Apr. 2017.

[11] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in Proc. 6th Int. Conf. Learning Representations (ICLR), Vancouver, BC, Canada, May 2018.

[12] Z. Liu, Y. Dou, P. S. Yu, Y. Deng, and H. Peng, "Alleviating the inconsistency problem of applying graph neural network to fraud detection," in Proc. 43rd Int. ACM SIGIR Conf. Research and Development in Information Retrieval, Virtual Event, China, Jul. 2020, pp. 1569–1572, doi: 10.1145/3397271.3401253.

[13] K. Salah, M. H. U. Rehman, N. Nizamuddin, and A. Al-Fuqaha, "Blockchain for AI: Review and open research challenges," IEEE Access, vol. 7, pp. 10127–10149, Jan. 2019, doi: 10.1109/ACCESS.2019.2893554.

[14] L. Yogarajan, M. Masukujjaman, M. H. Ali, N. Khalid, L. H. Osman, and S. S. Alam, "Exploring the hype of blockchain adoption in agri-food supply chain: A systematic literature review," Agriculture, vol. 13, no. 6, art. no. 1173, Jun. 2023, doi: 10.3390/agriculture13061173.

[15] S. Motie and B. Raahemi, "Financial fraud detection using graph neural networks: A systematic review," Expert Syst. Appl., vol. 240, art. no. 122156, Apr. 2024, doi: 10.1016/j.eswa.2023.122156.

[16] J. Ernstberger, S. Chaliasos, G. Kadianakis, S. Steinhorst, P. Jovanovic, A. Gervais, B. Livshits, and M. Orrù, "zk-Bench: A toolset for comparative evaluation and performance benchmarking of SNARKs," Cryptology ePrint Archive, Paper 2023/1503, 2023. [Online]. Available: https://eprint.iacr.org/2023/1503.

[17] C. Melo, G. Gonçalves, F. A. Silva, and A. Soares, "Performance modeling and evaluation of Hyperledger Fabric: An analysis based on transaction flow and endorsement policies," in Proc. 2024 IEEE Symp. Computers and Communications (ISCC), Paris, France, Jun. 2024, pp. 1–6.

[18] M. Shuaib, N. H. Hassan, S. Usman, S. Alam, N. A. Abu Bakar, and N. Maarop, "Performance evaluation of DLT systems based on Hyperledger Fabric," in Proc. 2022 4th Int. Conf. Smart Sensors and Application (ICSSA), Kuala Lumpur, Malaysia, Jul. 2022, pp. 70–75.

[19] S. Negi, "A blockchain technology for improving financial flows in humanitarian supply chains: Benefits and challenges," J. Humanitarian Logistics Supply Chain Manage., vol. 15, no. 2, pp. 154–174, 2025, doi: 10.1108/JHLSCM-10-2023-0099.